

House Price Prediction Using Machine Learning

PROJECT OVERVIEW

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics such as area, number of bedrooms, bathrooms, parking availability, furnishing status, and other housing features. The Housing Prices Dataset from Kaggle was used for the analysis.

DATA PREPARATION

The dataset was loaded using Pandas and examined for missing values, duplicates, and data quality issues. Duplicate records were removed, and categorical variables such as furnishing status, main road access, air conditioning, and preferred area were converted into numerical form using one-hot encoding. This preprocessing step ensured that the machine learning algorithms could effectively utilize all available information.

MODEL DEVELOPMENT

Two regression models were trained and evaluated:

- **Linear Regression**
- **Random Forest Regressor**

The dataset was divided into training and testing sets using an 80:20 ratio. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

MODEL PERFORMANCE & CODE INTEGRATION

To provide a more dynamic and interactive approach to analyzing results beyond static outputs, user interactive slicing features were implemented into the notebook using ipywidgets. This allows real-time rendering of performance metrics, data frames, and scatter plots based on cross-filtering choices like furnishing status or air conditioning presence, simulating a Power BI dashboard experience inside the Jupyter environment.

KEY INSIGHTS & SUMMARY

- **Most Influential Features:** Looking at our **correlation heatmap**, **house size (area)** and **air conditioning** have the strongest connection to higher prices, with extra bathrooms also playing a major role.

- **Model Accuracy:** In plain terms, the models are quite reliable. They won't get the price exactly perfect every single time, but they accurately capture about 60–65% of the overall price trends.
- **Surprising Discovery:** Interestingly, features like having a basement or being near a main road didn't matter nearly as much as just having more physical space and climate control.
- **Business Recommendation:** For a real estate company, the best move is to focus on properties with large layouts or flip houses by installing central AC. The data shows these specific traits bring in the highest return on investment.